

MID

Course: Data Structures (CS101)

Set: B | Time: 2 Hours | Max Marks: 60 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which of the following is NOT listed as one of the seven core design problems in the initial problem breakdown for Lexi? [5 Marks]
- a) Embellishing the UI
 - b) Database Connectivity
 - c) Spelling Checking and Hyphenation
 - d) Document Structure
2. What is the primary role of the Compositor class hierarchy within the context of Lexi's formatting problem? [5 Marks]
- a) To define the document's physical hierarchy of Glyphs.
 - b) To encapsulate different algorithms for breaking text into lines and columns.
 - c) To provide an interface for creating platform-specific UI widgets.
 - d) To manage user commands for undo and redo operations.
3. In Lexi's internal representation of document structure, what is the fundamental purpose of the Glyph abstract class? [5 Marks]
- a) It defines the interface for creating specific scroll bars or buttons.
 - b) It serves as the common base class for all elements, from individual characters to complex structural components.
 - c) It encapsulates the low-level, platform-specific operations for windowing systems.
 - d) It provides methods for dynamically adding visual embellishments to UI components.
4. What is a key characteristic of an embellishment (like Border or Scroller) when implemented using the Decorator pattern in Lexi? [5 Marks]
- a) It is a primitive element that cannot contain other glyphs.
 - b) It defines a low-level, platform-specific interface for windowing.
 - c) It is itself a Glyph that encloses another Glyph component and forwards requests.
 - d) It is an abstract factory responsible for creating families of related UI objects.
5. Which two design patterns are specifically mentioned as being combined to enable analytical operations like spelling checking and hyphenation, abstracting traversal and encapsulating analysis without altering Glyph classes? [5 Marks]
- a) Abstract Factory and Bridge
 - b) Strategy and Command
 - c) Iterator and Visitor
 - d) Composite and Decorator
6. The Bridge pattern is applied to Lexi's windowing abstractions primarily to achieve which of the following? [5 Marks]

- a) Provide a uniform mechanism for executing user commands from menus or buttons.
- b) Decouple the windowing abstraction from its platform-specific implementation, allowing independent variation.
- c) Encapsulate the formatting algorithm to allow easy swapping of speed and quality trade-offs.
- d) Support multiple look-and-feel standards without hard-coding class names throughout the application.

SECTION A: SHORT ANSWERS

7. Explain how Lexi uses the Composite pattern to solve the "Document Structure" problem, specifically addressing how it captures the physical hierarchy and treats text and graphics uniformly. **[6 Marks]**
8. Describe the mechanism by which the Decorator pattern in Lexi allows visual embellishments like borders and scroll bars to be added dynamically without creating a "combinatorial explosion of subclasses." **[6 Marks]**
9. Outline the specific roles of the GUIFactory abstract class and its concrete subclasses (e.g., MotifFactory, PMFactory) in supporting multiple look-and-feel standards within Lexi. **[6 Marks]**
10. Explain how the Bridge pattern separates Lexi's windowing abstraction from its implementation, enabling them to vary independently. **[6 Marks]**
11. Describe the mechanism by which the Command pattern provides a uniform approach for user operations and facilitates multi-level undo/redo functionality in Lexi. **[6 Marks]**

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